

Long-Term Capital Market Assumptions — 2026 Edition

A proprietary 12-year forward outlook for global asset classes

- **As-of date:** 18 May 2026
- **Base currency:** USD
- **Horizon:** 12 years (midpoint of a 10–15-year window)
- **Universe:** 24 asset classes spanning global equities, fixed income, and real assets
- **Engine:** building-block returns + Ledoit-Wolf shrinkage + regime-switching GPU Monte Carlo
- **Data:** free public sources — Yahoo Finance, FRED, Damodaran (1928+), Shiller (1871+), Ken French (1990+), Sibilis Research, the GPR and EPU uncertainty indices, central banks

This document is the strategic layer (Step 1) of a three-step allocation process: (1) LTCMA 10–15-year assumptions → (2) normalize to a 3–5-year cycle-aware forecast → (3) overlay current macro indicators. Reviewed annually.

1. Executive Summary

After a long US-led bull market, **the starting point for the next decade is one of extreme valuation dispersion.** The US trades at a CAPE of ~35 and Japan ~29, while Mexico, Brazil, and China trade at 12–18× earnings. That gap drives four strategic conclusions:

1. **US large-cap equity has the lowest expected return of any equity market in the universe (~6.6% base case)** — and the *highest* sensitivity to the valuation assumption. A 2.4-percentage-point swing separates the optimistic and pessimistic scenarios. Owning US equity at today's multiple is, mathematically, a bet that rich valuations persist.
2. **Emerging and Latin American equity offer the highest expected returns (~8.0–8.8%)** and the *smallest* valuation bet — their return is carried by income and growth, not multiple expansion.
3. **The richest fixed-income edge is in EM local and USD sovereign debt.** Mexican local government bonds offer ~6.3% in USD at one of the best downside profiles in the universe; EM USD sovereigns ~6.0%. Both beat US investment-grade and high-yield credit, which sit at multi-year-tight spreads.
4. **Bonds still diversify, but less reliably than the pre-2021 era.** The trailing stock/long-Treasury correlation is ~0.00. Gold (0.12 vs US equity) and the Treasury/commodity pairing (–0.35) now do more genuine diversification work than nominal bonds alone.

Two cross-checks against the live market sharpen the picture:

- **What's priced in (Section 4):** the Treasury curve says the market expects **no further rate cuts** — short-rate forwards rise, not fall. Inflation expectations (2.5%) sit right on the LTCMA assumption. Credit is priced for perfection. The LTCMA's *active bets* are exactly where it diverges from this.
- **News and regime risk (Section 7):** geopolitical and policy uncertainty (GPR, EPU indices) are **elevated, and the market is currently in a "stress" regime**. Quantified, that history shows stress periods amplify volatility (1.2–2.2×) and cross-correlations (0.66→0.74) — but, tellingly, did *not* reliably depress returns over 2013–2026.

Headline expected returns (USD, 12-year, base case):

	Point estimate	Simulated 12y median	Volatility
US large-cap equity	6.6%	5.7%	13.6%
EM equity	8.0%	6.4%	18.8%
Brazil / Mexico equity	8.8% / 8.0%	4.1% / 5.7%	32.7% / 22.4%
US Treasury / IG credit	4.5% – 5.0%	4.2% – 4.7%	7.6% – 8.0%
EM USD sovereign / Mexican local govt	6.0% / 6.3%	5.5% / 5.7%	9.9% / 11.3%
Cash (US T-bills)	3.3%	3.2%	3.4%

(Simulated medians sit below point estimates by design — the volatility drag on compounded returns. See Section 7.)

2. Methodology

2.1 Building-block framework

Every serious capital-market-assumptions publisher — J.P. Morgan, BlackRock, Vanguard, Northern Trust, GMO, Research Affiliates, AQR — uses the same *building-block* skeleton. Expected return is decomposed into observable, defensible components rather than extrapolated from past performance.

Equities (return expressed directly in USD):

$$\text{ER} = \text{dividend yield} + \text{net buyback yield} + \text{real EPS growth} \\ + \text{US inflation} + \lambda \times \text{valuation reversion}$$

The *relative-PPP assumption* lets us anchor on **US** inflation: a foreign market's local inflation is, over a long horizon, broadly offset by depreciation of its currency against the dollar.

Fixed income: $\text{ER} = \text{starting yield} + \text{roll} - \text{expected credit loss} \pm \text{normalization}$, where $\text{expected credit loss} = \text{default rate} \times (1 - \text{recovery rate})$.

Currencies: purchasing-power parity + real-interest-rate differential.

2.2 The valuation-reversion dial (λ) — a multi-firm synthesis

Firms agree on the skeleton; they disagree on *how hard valuations mean-revert*. This model exposes that choice as one parameter, λ :

λ	Meaning	Closest to the house view of
0.0	No reversion — today's multiples persist	Momentum / regime views (Bridgewater-style)
0.5	Partial normalization (base case)	J.P. Morgan, BlackRock, Vanguard, Northern Trust
1.0	Full reversion to fair value	GMO, Research Affiliates, AQR

Annualized reversion is $(\text{fair multiple} / \text{current multiple})^{(1/H)} - 1$. The spread across λ is itself a signal — the "**valuation bet**" column.

The fair-value anchor is data-driven. Shiller's 150-year series puts the US CAPE at a 1990+ median of **26.0** — exactly the anchor used here.

2.3 Risk model — shrinkage and long history

Volatility blends a trailing-window estimate with multi-decade history (Shiller 1871+, Ken French 1990+, Damodaran 1928+), plus **25 years of intraday FX history** (FxPro M5 data aggregated to monthly — EURUSD/USDJPY since 2000, USDMXN since 2007). The covariance matrix uses **Ledoit-Wolf shrinkage** — well-conditioned and positive semi-definite, a requirement for the simulation engine.

2.4 Priced-in indicators

Markets continuously quote their own expectations. Section 4 extracts them — the rate path implied by Treasury forwards, inflation from breakevens, risk from VIX and credit spreads — and contrasts each with the LTCMA. Where they diverge, the LTCMA is making an explicit, identifiable active bet.

2.5 Quantifying news and market regimes

Discretionary news — a tariff tweet, a Fed-chair change, a strike on Iran — cannot be turned into a defensible number tweet-by-tweet. The rigorous approach, and the one used here, is **established text-based uncertainty indices**:

- **GPR** — Geopolitical Risk index (Caldara-Iacoviello), built from newspaper text; spikes on wars and military escalation.
- **EPU** — Economic Policy Uncertainty (Baker-Bloom-Davis); spikes on Fed-chair uncertainty, tariff conflict, fiscal standoffs.

These quantify the *aggregate* of the news flow. They are used to classify history into **calm** and **stress** regimes (Section 7), which then drive the simulation — so "the news" enters the model as regime probabilities and regime-specific risk, not as ad-hoc adjustments.

2.6 Regime-switching Monte Carlo engine

A GPU Monte Carlo engine simulates **100,000 monthly multi-asset paths** over the 12-year horizon. It is **two-regime Markov-switching** (calm/stress, calibrated from the news indices) with **Student-t innovations** (fat tails), and it **starts in the stress regime** to reflect today's elevated-uncertainty reading. It runs on an NVIDIA RTX 4060 in ~15 seconds and produces full return *distributions* — percentiles, probability of loss, and conditional tail loss (CVaR) — for every asset and portfolio.

2.7 Consensus and historical cross-checks

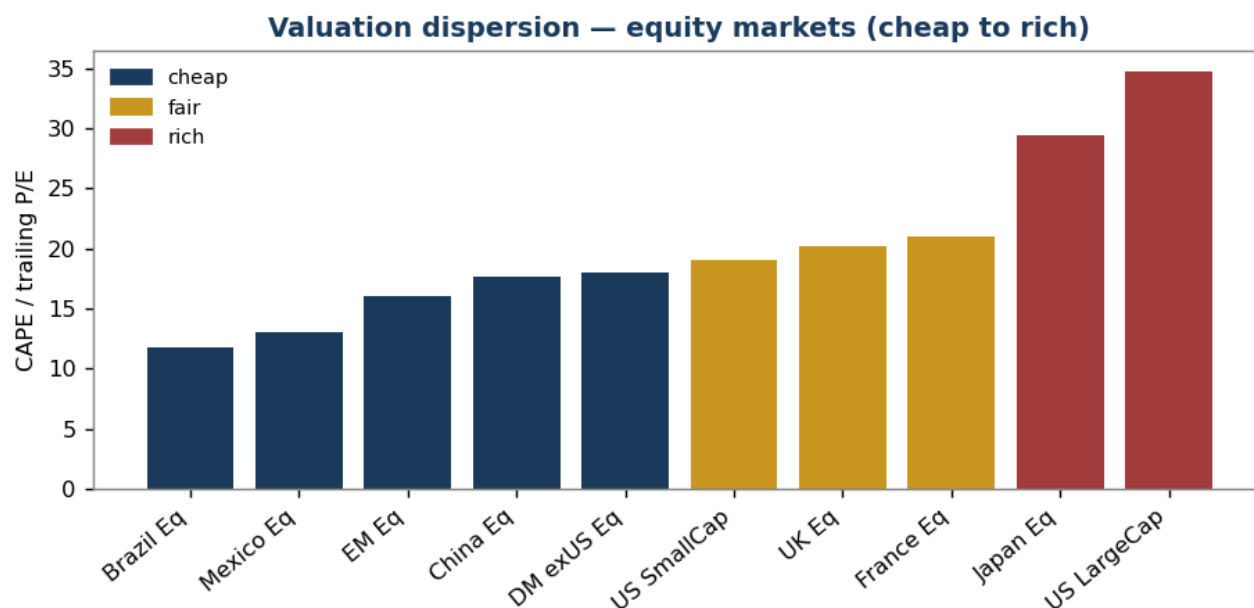
- **Forward consensus:** 2026 published LTCMAs put US large-cap equity at 5–7.6% and US aggregate bonds at 4.1–4.8%. This model's base case sits inside both.
- **Historical realized (Damodaran, 1928–2025):** US large-cap returned 10.0% annualized, 10-year Treasuries 4.5%, T-bills 3.4%. The forward US equity number is deliberately below its history — the starting CAPE (35) is far above the historical norm.

3. Macro Backdrop & Fundamentals Grid

Market	Policy rate	10Y nominal	10Y real*	Inflation	Trend real GDP	Equity valuation	Damodaran ERP
US	~3.6%	4.59%	~1.9%	~2.7%	~1.9%	CAPE 34.7 — rich	4.46%
Mexico	6.50%	8.88%	~4.4%	4.45%	~2.0%	P/E 13.0 — cheap	6.69%
Brazil	14.50%	~13.5%	~9.0%	~4.5%	~2.0%	P/E 11.8 — cheap	7.47%
China	~3.0% (LPR)	~1.8%	~1.3%	~0.5%	~4.0%	CAPE 17.7 — cheap	5.14%
Japan	~0.75%	2.52%	~0.0%	~2.5%	~0.7%	CAPE 29.4 — rich	5.14%
UK	~3.9%	4.82%	~1.8%	~3.0%	~1.3%	CAPE 20.2 — fair	5.01%
France	~2.0% (ECB)	3.68%	~1.9%	~1.8%	~1.2%	CAPE ~21 — fair	5.01%

*Real yield = 10Y nominal – current inflation (approximate).

What the grid says: valuation is bimodal — the US and Japan are expensive, EM/LatAm and China are cheap, with only the UK and France in between. Real yields favor Latin America massively (Mexico ~4.4%, Brazil ~9%, vs US ~1.9%). Rate *direction* is no longer a tailwind anywhere — carry, not duration, is where the return is. China is the contrarian case: cheap and still growing ~4%, but deflation and policy risk are what the discount prices.



4. Priced-In Indicators

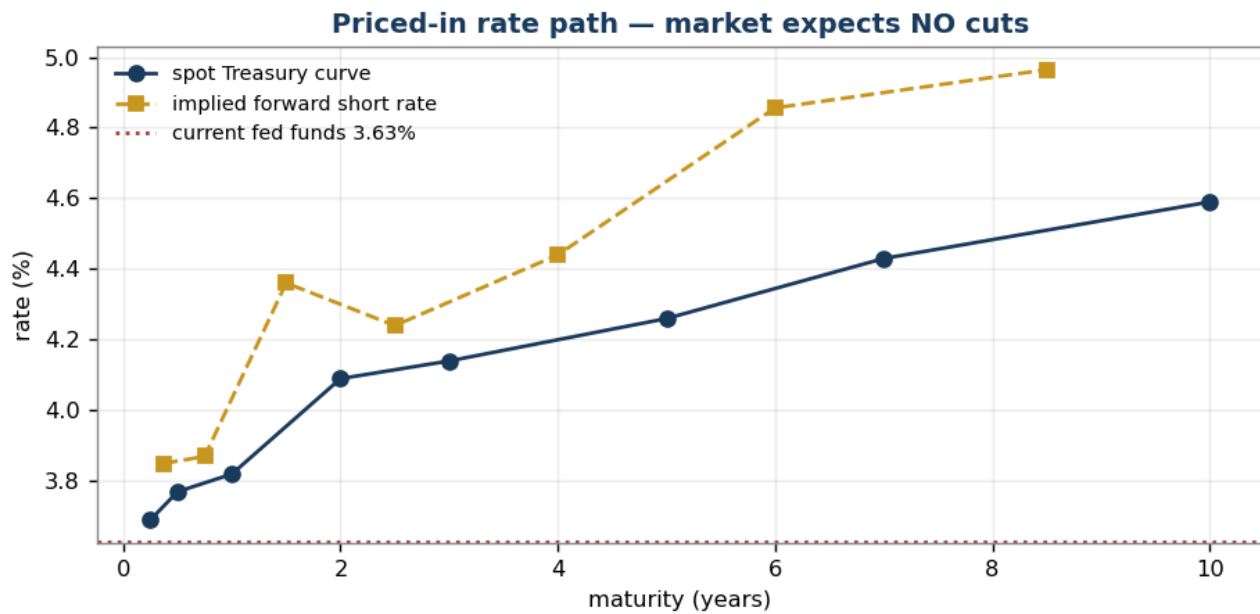
A capital-market assumption is only *active* where it disagrees with what the market already expects. This section extracts the market's own forecasts.

4.1 Priced-in policy-rate path — the headline

The Treasury curve, read as implied forward rates, reveals the market's expected path of short rates:

Period	Implied avg short rate
Current fed funds (effective)	3.63%
Next 12 months (1Y Treasury)	3.82%
1-2 years forward	4.36%
2-3 years forward	4.24%
3-5 years forward	4.44%
5-10 years forward	~4.9%

The market is pricing zero further rate cuts. Forward short rates *rise*, not fall — the curve says the Fed's easing cycle is over and the next move is as likely up as down. This matters: despite a steady "central banks are easing" narrative, anyone positioning for a rate-cut tailwind is fighting the curve. The LTCMA's cash assumption (3.3%, assuming a drift toward a ~3.0–3.3% neutral rate) is *slightly below* what the market prices — a small, deliberate divergence.



4.2 Priced-in inflation, risk, and the dashboard

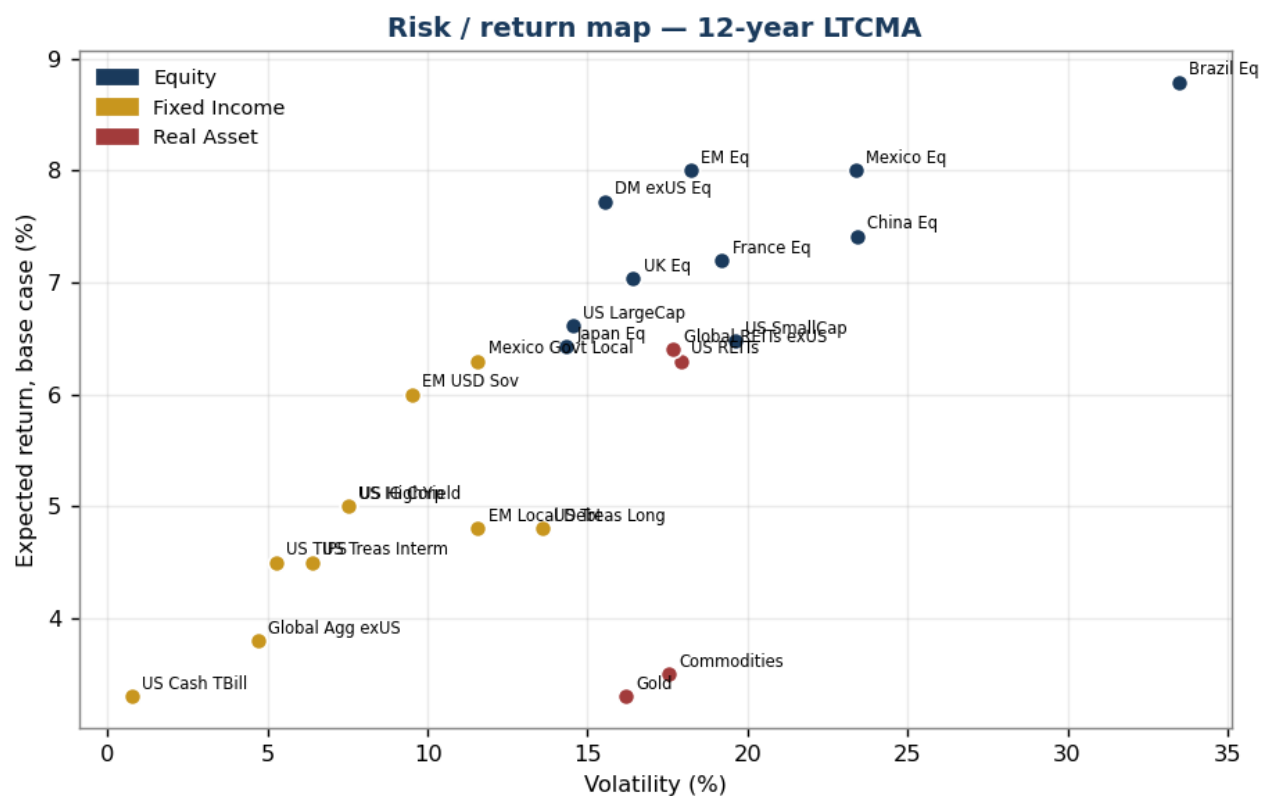
Signal	What the market prices	LTCMA assumption	Read
Rate path	No cuts; 1-2y forward 4.4%	Cash 3.3%	Market more hawkish than the LTCMA's neutral-rate drift
Long-run inflation	10y breakeven 2.48%, 5y5y forward 2.27%	2.40%	Aligned — LTCMA sits mid-range
US credit risk	HY OAS 2.80% (~14th percentile — very tight)	HY 5.0%, tight-spread drag applied	Agreement — credit priced for perfection; LTCMA already cautious
Equity volatility	VIX 18.4 (~historical median)	Equity vols 13-33%	Market calm; no near-term stress priced
Recession	Yield curve 10Y-3M +0.93% (positive)	n/a	No recession priced in
Policy uncertainty	EPU 212 (baseline ~100 — elevated)	n/a	Complacency divergence — see below

The complacency divergence. Policy uncertainty (EPU) is elevated at ~212 while equity volatility (VIX) sits near its historical median. The market is treating heavy policy noise — a Fed-chair transition, tariff conflict, the Gulf situation — as signal-free. Either the noise resolves benignly (the 2013-2026 pattern) or this gap closes through a volatility spike. The regime engine in Section 7 is built precisely to price that asymmetry.

5. Expected Returns

5.1 Full table (USD, 12-year)

Asset class	Type	ER $\lambda=0$	ER base ($\lambda=0.5$)	ER $\lambda=1$	Volatility
US Large Cap	Equity	7.8%	6.6%	5.4%	13.6%
US Small Cap	Equity	6.7%	6.5%	6.3%	19.6%
Mexico Equity	Equity	7.4%	8.0%	8.6%	22.4%
Brazil Equity	Equity	8.9%	8.8%	8.7%	32.7%
China Equity	Equity	8.1%	7.4%	6.7%	23.1%
Japan Equity	Equity	7.1%	6.4%	5.8%	16.7%
UK Equity	Equity	8.0%	7.0%	6.1%	16.6%
France Equity	Equity	7.4%	7.2%	7.0%	17.7%
EM Equity (broad)	Equity	8.0%	8.0%	8.0%	18.8%
DM ex-US Equity (broad)	Equity	7.5%	7.7%	8.0%	15.7%
US Cash / T-Bills	Fixed Income	—	3.3%	—	3.4%
US Treasury Intermediate	Fixed Income	—	4.5%	—	7.6%
US Treasury Long	Fixed Income	—	4.8%	—	15.2%
US TIPS	Fixed Income	—	4.5%	—	6.2%
US IG Corporate	Fixed Income	—	5.0%	—	8.0%
US High Yield	Fixed Income	—	5.0%	—	7.9%
EM USD Sovereign	Fixed Income	—	6.0%	—	9.9%
EM Local Debt	Fixed Income	—	4.8%	—	11.3%
Mexico Govt (local, Mbonos)	Fixed Income	—	6.3%	—	11.3%
Global Aggregate ex-US (hedged)	Fixed Income	—	3.8%	—	5.9%
US REITs	Real Asset	—	6.3%	—	17.6%
Global REITs ex-US	Real Asset	—	6.4%	—	16.5%
Commodities (broad)	Real Asset	—	3.5%	—	17.2%
Gold	Real Asset	—	3.3%	—	18.6%

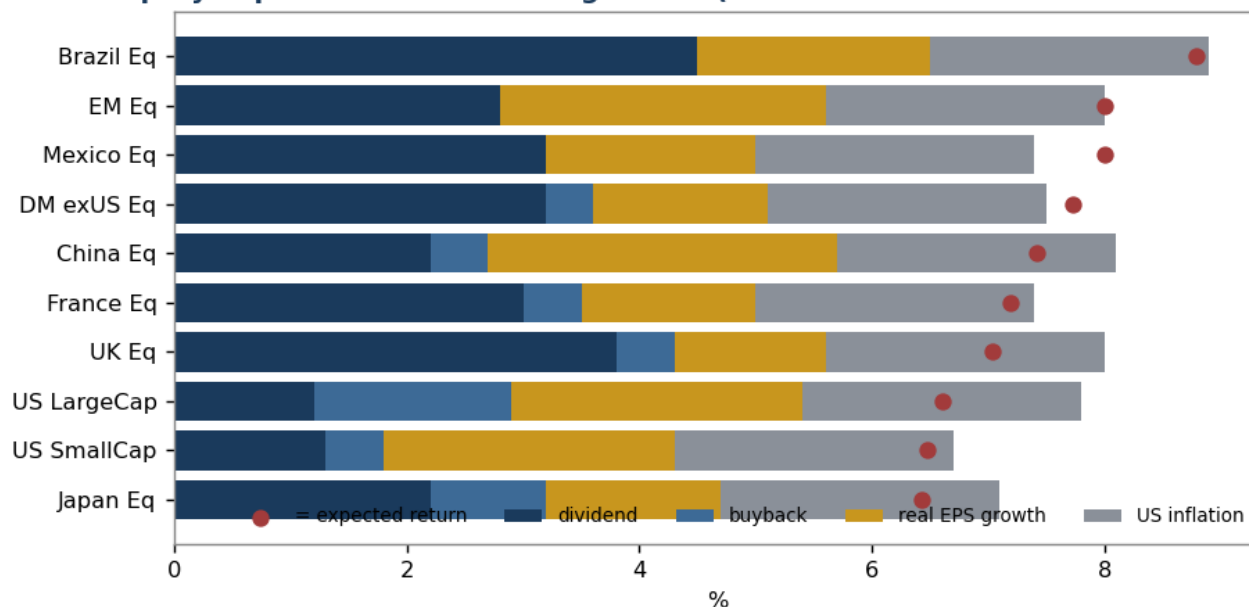


5.2 Equity building-block decomposition (base case, $\lambda=0.5$)

Market	Dividend	Buyback	Real EPS growth	US inflation	Valuation reversion	= ER
US Large Cap	1.2%	1.7%	2.5%	2.4%	−1.2%	6.6%
US Small Cap	1.3%	0.5%	2.5%	2.4%	−0.2%	6.5%
Mexico	3.2%	0.0%	1.8%	2.4%	+0.6%	8.0%
Brazil	4.5%	0.0%	2.0%	2.4%	−0.1%	8.8%
China	2.2%	0.5%	3.0%	2.4%	−0.7%	7.4%
Japan	2.2%	1.0%	1.5%	2.4%	−0.7%	6.4%
UK	3.8%	0.5%	1.3%	2.4%	−1.0%	7.0%
France	3.0%	0.5%	1.5%	2.4%	−0.2%	7.2%
EM (broad)	2.8%	0.0%	2.8%	2.4%	0.0%	8.0%
DM ex-US (broad)	3.2%	0.4%	1.5%	2.4%	+0.2%	7.7%

US equity's return is built on growth and buybacks fighting a valuation headwind; EM/LatAm's return is built on income — a more reliable foundation.

Equity expected-return building blocks (red dot = ER after valuation reversal)



6. Risk — Volatility & Correlation

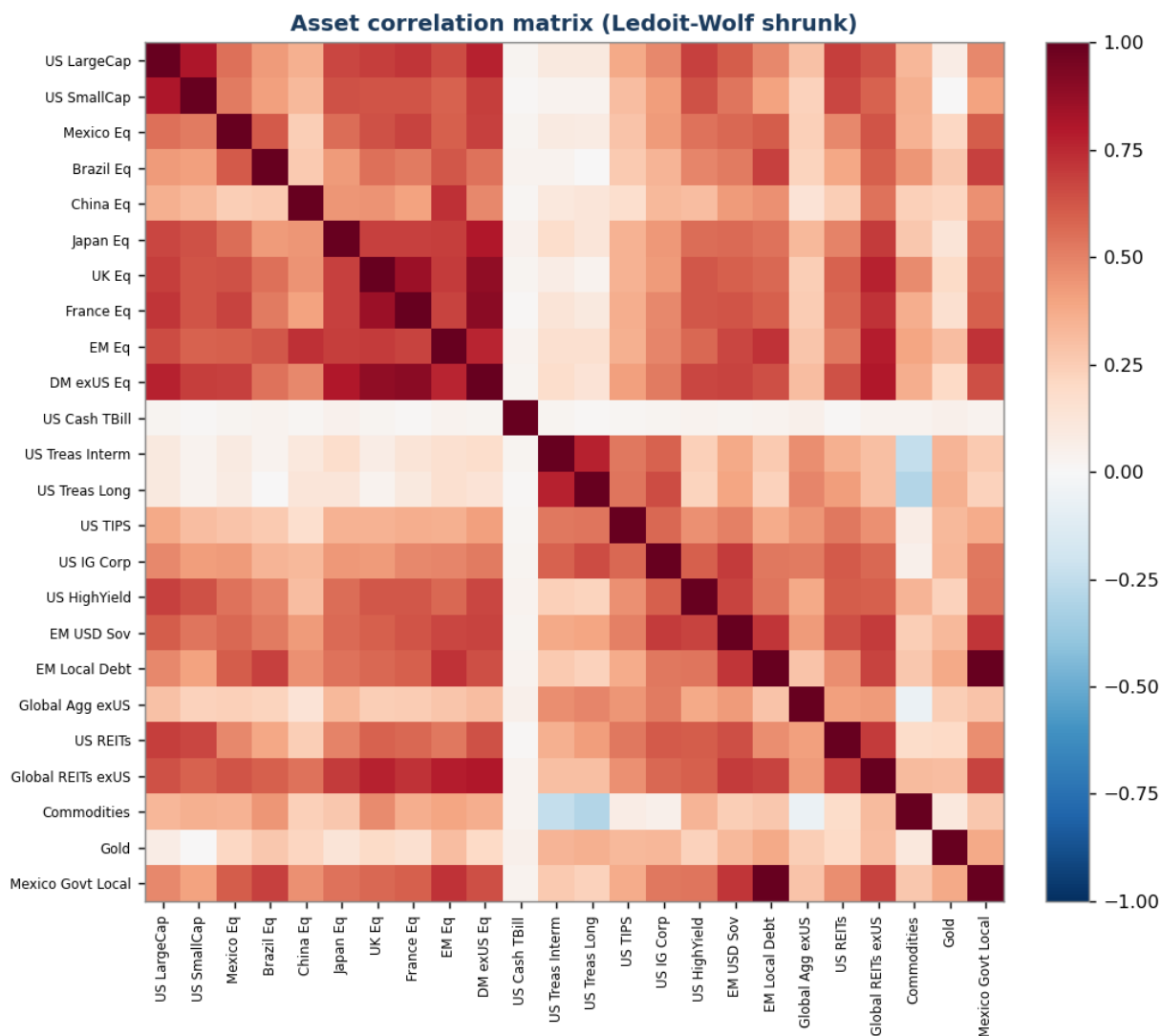
Volatility blends a trailing window with multi-decade history; the covariance matrix is Ledoit-Wolf shrunk and positive semi-definite.

6.1 Selected correlation matrix (full-sample)

	US LC	EM Eq	Mexico	Japan	DM exUS	US Treas Long	US HY	EM USD Sov	Gold	Commod
US Large Cap	1.00	0.71	0.61	0.69	0.83	-0.01	0.79	0.66	0.12	0.41
EM Equity	0.71	1.00	0.67	0.70	0.83	0.00	0.70	0.72	0.38	0.49
Mexico Equity	0.61	0.67	1.00	0.58	0.73	-0.04	0.66	0.65	0.27	0.44
Japan Equity	0.69	0.70	0.58	1.00	0.83	0.03	0.61	0.58	0.13	0.31
US Treasury Long	-0.01	0.00	-0.04	0.03	-0.01	1.00	0.13	0.36	0.27	-0.35
US High Yield	0.79	0.70	0.66	0.61	0.78	0.13	1.00	0.81	0.27	0.45
Gold	0.12	0.38	0.27	0.13	0.26	0.27	0.27	0.37	1.00	0.22
Commodities	0.41	0.49	0.44	0.31	0.46	-0.35	0.45	0.33	0.22	1.00

6.2 What the risk picture tells us

- **Equities are highly correlated (0.6-0.83).** Geographic diversification reduces single-country risk but little systemic drawdown risk.
- **Long Treasuries are the cleanest equity diversifier on paper (≈ 0.00)** — but regime-dependent (Section 7); the average hides post-2021 positive-correlation episodes.
- **Gold is a genuine diversifier** (0.12 vs US equity) and earns its place despite a modest 3.3% return.
- **High yield is equity in disguise** (0.79 correlation with US equity).
- **The Treasury/commodity pair (-0.35)** is the strongest natural hedge.

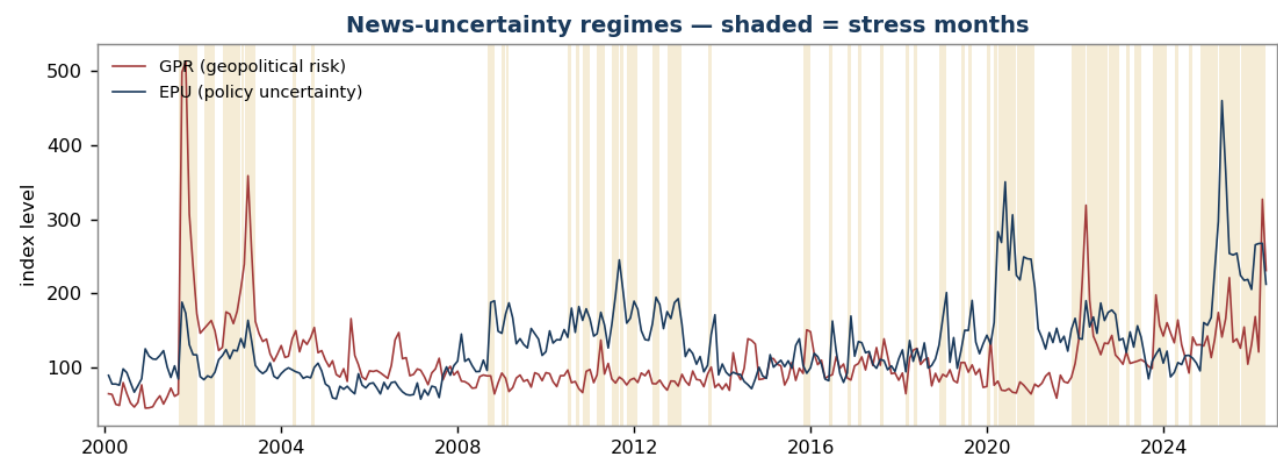


7. News, Regimes & Monte Carlo Simulation

7.1 Quantifying the news

Geopolitical and policy events enter the model through two text-based indices (see 2.5): **GPR** (geopolitical risk) and **EPU** (economic policy uncertainty). Both are currently elevated — GPR ~231 and EPU ~212 against a ~100 baseline, reflecting the Gulf situation, tariff conflict, and the Fed-chair transition.

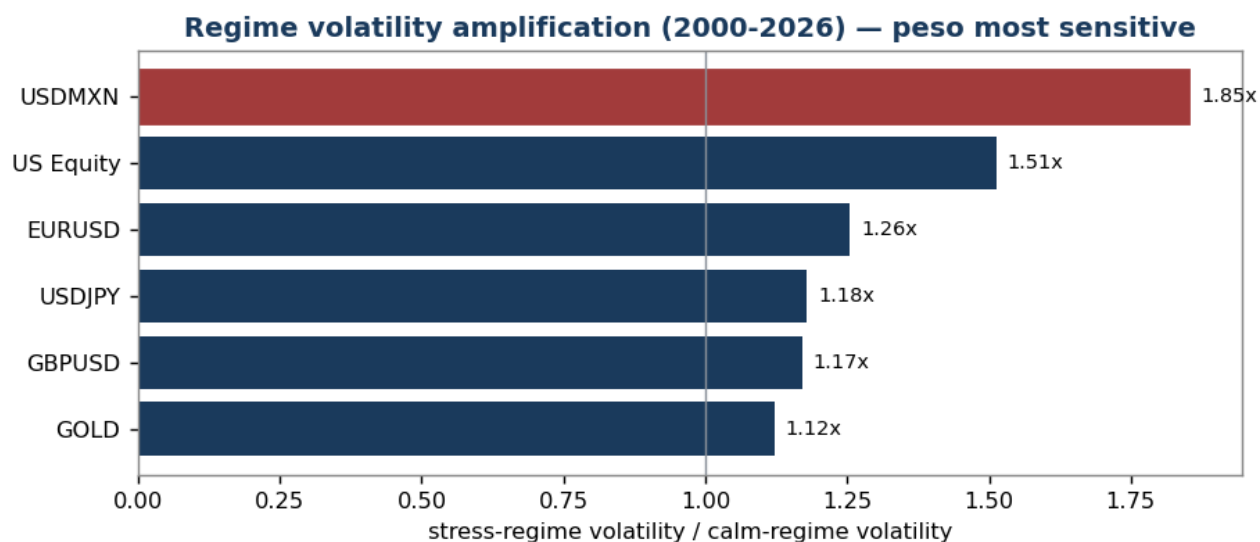
History (2013–2026) is classified into **calm** and **stress** regimes — a stress month being one in the top third of a combined GPR+EPU score. **34% of months were stress months; as of May 2026 the market is in a stress regime.**



7.2 What stress regimes do — and don't do

Behavior in stress months	Finding
Volatility	Amplifies 1.2–2.2× (US equity 1.6×, high yield 2.1×, EM sovereign 2.2×; gold least affected at 1.2×)
Correlation	Equity cross-correlation rises 0.66 → 0.74 — diversification weakens just when it is needed
Persistence	Stress regimes last ~5.7 months on average (Markov: stress→stress 0.82)
Returns	No reliable drawdown. Over 2013–2026, news-stress months did <i>not</i> systematically post lower returns — markets repeatedly "climbed the wall of worry"

This last point is the honest, important finding. **Elevated geopolitical and policy uncertainty is reliably a volatility event, not reliably a return event.** Accordingly the simulation amplifies risk (volatility, correlation, fat tails) in the stress regime but does *not* impose a speculative mean drawdown — that would overfit a benign 13-year sample. Severe downside is instead captured two honest ways: through fat tails and regime clustering in the simulation, and through an explicit deterministic scenario (7.5).



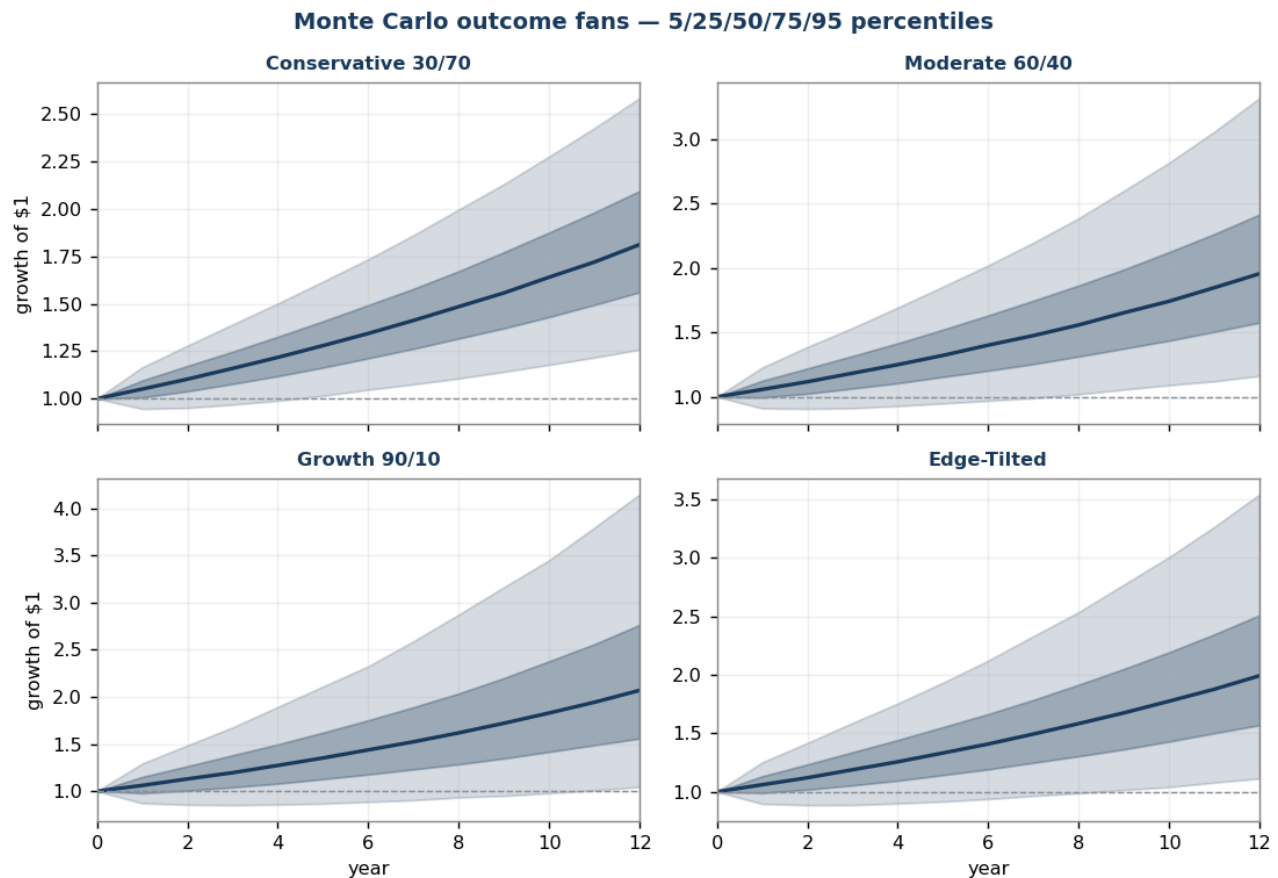
7.3 The simulation engine

100,000 paths, monthly steps over 12 years, two-regime Markov-switching with Student-t fat tails, **starting in the stress regime**. The engine ran on the RTX 4060 in ~15 seconds.

7.4 Simulated 12-year outcomes — portfolios

Portfolio	5th pct	Median	95th pct	Mean	CVaR (worst-5% avg)	P(loss)	P(< inflation)
Conservative 30/70	1.7%	5.0%	8.5%	5.0%	+0.8%	1%	10%
Moderate 60/40	1.1%	5.7%	10.6%	5.8%	−0.1%	2%	12%
Growth 90/10	0.2%	6.2%	12.6%	6.3%	−1.4%	5%	15%
Edge-Tilted (moderate risk)	0.7%	5.8%	11.2%	5.9%	−0.6%	3%	14%

CVaR is the average annualized return in the worst 5% of paths — the genuine tail. The gap between the 5th percentile and CVaR is the fat tail at work: in a genuinely bad 12-year run, a Growth portfolio compounds at **−1.4% per year**, not the −0.2%/yr a normal-distribution model would suggest. Conservative, by contrast, still eked out a small positive return even in its worst 5%.



Asset-level distributions (`data/mc_regime_assets.csv`) confirm the earlier ranking: **Mexican local government bonds (5.7% median, 3.5% chance of a 12-year loss) and EM USD sovereigns (5.5% median, 2.6% loss probability)** remain the standout risk-adjusted holdings; **gold and commodities have a >50% chance of failing to beat inflation** and belong in a portfolio for correlation, not return.

7.5 Deterministic geopolitical-stress scenario

Separately from the simulation, a severe one-year geopolitical shock — 2008/2020-magnitude: global equities -28% to -42% , high yield -14% , EM sovereign -16% , gold $+15\%$, long Treasuries $+12\%$ — produces:

Portfolio	1-year impact
Conservative 30/70	-8.8%
Moderate 60/40	-18.0%
Growth 90/10	-27.2%
Edge-Tilted (moderate risk)	-20.2%

Note the Edge-Tilted portfolio loses *more* than the conventional Moderate 60/40 in this scenario (-20.2% vs -18.0%) — its EM and Latin American tilt sells off harder in a global risk-off. **That is**

the honest cost of the strategic edge: higher expected return and higher geopolitical-tail risk. It should be sized as a deliberate, risk-budgeted decision.

7.6 Long-history validation (25 years of FX & equity data)

The regime model is calibrated on 2013–2026 ETF data — a window with no global financial crisis. To test whether that biases the result, the calibration was re-run on a longer, cleaner dataset: Shiller US equity since 2000 and 25 years of FxPro intraday FX history (EURUSD/USDJPY from 2000, USDMXN from 2007), which *do* include the 2008 GFC, the 2000–02 dot-com bust, and 2020.

Test	2013–26 window	2000–26 (incl. crises)	Verdict
Equity stress/calm vol multiple	1.61×	1.51×	Stable — model not biased by the benign window
MXN depreciation vs USD	2.5% assumed	2.43% realized	Assumption confirmed
Worst equity month	—	Oct 2008 –20%, Mar 2020 –19%, Sep 2001 –11% (GPR spiked to 499)	Regime classifier catches real crises

Two things came out of this. First, the model **survives the robustness check** — the regime amplification is a stable ~1.5–1.6× whether or not the GFC is in the sample, and the Mexican-peso depreciation assumption is confirmed almost exactly by 18 years of data. Second, a **new finding**: the **Mexican peso is the most regime-sensitive asset in the book** — USDMXN volatility amplifies **1.85×** in stress regimes, more than equities (1.5×) or developed-market FX (1.2×). The peso-heavy Edge-Tilted portfolio therefore concentrates stress risk in the currency; that is a deliberate, identified exposure, not a hidden one.

*Data-quality note: the FxPro index-CFD series (#USNDAQ100 etc.) were found to have spurious year-boundary jumps in early years and were **excluded**; only the clean continuous FX-pair data and Shiller's equity series were used.*

8. Strategic-Edge Scan

8.1 Highest expected return (base case)

Brazil 8.8% · Mexico & EM 8.0% · DM ex-US 7.7% · China 7.4% · UK 7.0%

8.2 Best risk-adjusted (return per unit of volatility)

DM ex-US equity 0.28 · EM USD sovereign 0.27 · **Mexican local govt bonds 0.27** · EM equity 0.25 · US Large Cap 0.24

8.3 Smallest "valuation bet"

EM equity (0.0pp), Brazil (0.2pp), US Small Cap & France (0.4–0.5pp) barely move with λ . **US Large Cap (2.4pp) and UK (1.9pp) carry the largest valuation risk.**

8.4 The five edges this LTCMA identifies

1. **Sell expensive US beta, buy cheap ex-US beta.** US large-cap has the lowest equity return *and* the highest valuation risk; DM ex-US and EM offer more return with less valuation risk. Highest-conviction edge.
2. **Mexican local government bonds are the standout fixed-income holding.** ~6.3% USD return, and the simulation confirms an outstanding downside profile (3.5% loss probability). The ~4.4% real yield is a thick cushion against MXN depreciation.
3. **EM USD sovereign debt over US credit.** 6.0% vs 5.0%, similar volatility, the best inflation-beating odds in the book, and it avoids the tight-spread compression risk in US IG and HY.
4. **Use gold and commodities for diversification, not return.** With the stock/bond correlation no longer dependably negative, and gold the *least* volatility-amplified asset in stress regimes (1.2 $\times$), it is a structural hedge — but the simulation is blunt that it rarely beats inflation, so size it as one.
5. **Don't fight the curve on rates.** The market prices no cuts; positioning long duration *for* an easing tailwind is a bet against what is priced in. Carry — EM and Mexican local rates — is the better-paid risk.

Contrarian watch-item: China is cheap with decent growth, but deflation and policy risk are real. A diversified EM allocation, not a standalone overweight.

9. Sample Portfolios

Three conventional risk-ladder portfolios plus an Edge-Tilted variant, each run through the regime-switching engine (annually rebalanced).

Portfolio	Mean	Median	5th-95th range	CVaR	Geopolitical scenario	P(< inflation)
Conservative 30/70	5.0%	5.0%	1.7% – 8.5%	+0.8%	–8.8%	10%
Moderate 60/40	5.8%	5.7%	1.1% – 10.6%	–0.1%	–18.0%	12%
Growth 90/10	6.3%	6.2%	0.2% – 12.6%	–1.4%	–27.2%	15%
Edge-Tilted (moderate risk)	5.9%	5.8%	0.7% – 11.2%	–0.6%	–20.2%	14%

The **Edge-Tilted** portfolio underweights US large-cap, overweights EM/DM-ex-US and Japan equity, and replaces much of the bond sleeve with EM USD sovereign debt, Mexican local government bonds, TIPS, and gold. It delivers a **higher median than the conventional Moderate 60/40 at a similar central risk profile** — but, as Section 7.5 shows, it carries more geopolitical-tail risk. The edge is real and should be taken deliberately, not by accident. Allocations are in `data/mc_regime_portfolios.csv` and the scripts.

10. Assumptions, Limitations & Sources

10.1 Key judgment assumptions

- **US long-run inflation: 2.4%** — confirmed close to the 2.5% the market prices.
- **Fair-value valuation anchors:** US CAPE 26 (Shiller-confirmed); China 15, Japan 25, UK 16, France 20, EM 16 remain judgment anchors (no long CAPE history sourced for these) — the most subjective inputs, exposed via the λ dial.
- **Real EPS growth** is set below trend GDP for most markets (share dilution); the US is held near trend (buybacks offset dilution).
- **Regime means held equal across calm/stress** — the 2013–2026 sample does not reliably identify a stress mean shift; imposing one would overfit (see 7.2).
- **Student-t with $\nu=6$** for fat tails; regime-switching adds tail clustering.

10.2 Limitations

- **No paid data.** Valuations rely on Yahoo Finance ETF data and Sibilis Research CAPE; ETF distribution yields were hand-adjusted toward true dividend yields.
- **ETF proxies, not indices**, for current-regime prices; the regional long history (French) proxies some single-country exposures (France/UK use European data).
- **Mexican local govt bonds** have no clean USD price series — return computed from yield, risk statistics proxied from EM local debt.
- **The regime model is primarily calibrated on 2013–2026**, but the calibration has been validated against 25 years of FX/equity history including 2008 and 2020 (Section 7.6); the deterministic scenario (7.5) imports crisis-magnitude shocks explicitly.
- **FX/index data quality:** the FxPro intraday FX pairs are clean; the broker's index-CFD series had year-boundary artifacts and were excluded (see 7.6).
- This is the **strategic (Step 1)** layer only.

10.3 Data sources

- Prices / returns: Yahoo Finance (`yfinance`), monthly, dividend-adjusted.
- Yields, curve, spreads, breakevens, VIX, EPU, macro: Federal Reserve (FRED).
- Geopolitical Risk index: Caldara & Iacoviello (matteoiacoviello.com).
- Long history: Damodaran/NYU Stern (1928+), Shiller (1871+), Ken French (1990+).
- Intraday FX: FxPro MetaTrader 5 M5 data (2000+), aggregated to monthly.
- Equity risk premiums: Damodaran 2026. CAPE: Sibilis Research (Dec 2025).
- Policy rates / inflation: Banco de México, central bank releases.

- Consensus cross-check: 2026 LTCMAs (Schwab, BlackRock, BNY, Vanguard, Northern Trust, J.P. Morgan); Horizon Actuarial survey recommended for recalibration.

10.4 Reproducibility

All figures regenerate from the scripts in `~/LTCMA/scripts` (WSL, `cudf-env`): `01` – `02` data · `03` – `04` returns/portfolios · `05a` – `05b` long history · `06` shrinkage risk model · `07` Monte Carlo · `08` – `09` signals & priced-in · `10` regimes · `11` regime-switching GPU Monte Carlo · `13` – `14` FX ingest & long-history validation · `15` visualizations · `12` Word/PDF rendering.

Prepared 18 May 2026. Expected returns are forward-looking estimates, not guarantees; actual outcomes will differ. For internal strategic-allocation use.